

# Hardware-Accelerated Orthogonal Polynomial Tensor Operators, Zero-Backpropagation Closed-Form Algebraic Solvers, and In-Situ Weight Surgery for Deep Neural Networks\*

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## Abstract

Standard deep neural networks and transformer feed-forward architectures (e.g., SwiGLU) rely heavily on high-dimensional multi-matrix projections optimized through iterative, compute-intensive gradient backpropagation (e.g., AdamW). This paradigm suffers from slow convergence, substantial GPU compute costs, and catastrophic forgetting during fine-tuning. In this work, we propose **Orthogonal Polynomial Tensor Operators** and a **Zero-Backpropagation Closed-Form Algebraic Solver** operating on an idempotent projection manifold ( $P^2 = P$ ). By mapping intermediate token activations into orthogonal Chebyshev polynomial bases  $T_k(x)$  and solving the parameter tensor via single-pass singular value decomposition (SVD) and regularized normal equations, our solver achieves a **240.9× acceleration** in convergence time over 150-epoch AdamW optimization while establishing exact algebraic idempotency ( $\|\Pi^2 - \Pi\|_F / \|\Pi\|_F = 4.70 \times 10^{-5}$ ). Furthermore, we formulate an **In-Situ Neural Weight Surgery** protocol with Residual Polynomial Adapters (ResPoly). We validate our method on **SmolLM2-135M-Instruct**, eliminating **50.0% of FFN parameters** (up to 39.76M parameters across 30 layers) in under 60 ms per layer with verified natural text coherence. The implementation is open-core under the **idempotent-poly** library on PyPI and verified on modern GPU accelerators.

## 1 Introduction

Modern Large Language Models (LLMs) dedicate over 60% of their parameters to feed-forward networks

(FFNs) [6, 8]. Current implementations combine up-projection, gate-projection, element-wise non-linear activations (SiLU/GELU), and down-projection [7]. Training or fine-tuning these structures demands substantial FLOP budgets and multi-pass gradient backpropagation.

To fundamentally bypass iterative backpropagation, we introduce an algebraic paradigm: expressing neural transformations as multi-rank orthogonal polynomial tensors whose optimal coefficients are derived analytically in closed form via idempotent subspace projections [2, 1].

## 2 Orthogonal Polynomial Tensor Algebra

**Definition 1** (Orthogonal Chebyshev Tensor Operator). *Let  $x \in [-1, 1]^D$  denote a normalized activation vector. The Chebyshev polynomials of the first kind [5] satisfy the recurrence:*

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x) \quad (1)$$

*For an input tensor  $X \in \mathbb{R}^{B \times D}$ , the polynomial feature map contracts with coefficient tensor  $\mathcal{C} \in \mathbb{R}^{D_{out} \times D_{in} \times (K+1)}$ :*

$$Y = \sum_{k=0}^K \mathcal{C}_{:, :, k} \cdot T_k(X) + b \quad (2)$$

**Theorem 1** (FFN Parameter Halving). *For a standard SwiGLU FFN with hidden dimension  $D$  and intermediate dimension  $D_{ff} = \frac{8}{3}D$ , total parameters equal  $3DD_{ff} = 8D^2$ . In contrast, an orthogonal Chebyshev tensor layer with degree  $K = 3$  has total parameters  $(K + 1)D^2 = 4D^2$ , yielding a theoretical **50.0% reduction** in FFN parameters.*

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\*Patent Notice: The orthogonal polynomial tensor operators, zero-backpropagation closed-form algebraic solvers, in-situ weight surgery algorithms, and PolyFormer architectures disclosed in this paper are protected under U.S. Patent Application No. 64/149,540 ("Patent Pending", Confirmation No. 1756, claiming priority under 35 U.S.C. §119(e) to U.S. Provisional Application No. 64/148,668).

### 3 Zero-Backpropagation Closed-Form Solver

Given calibration activations  $X \in \mathbb{R}^{N \times D}$  and target representations  $Y \in \mathbb{R}^{N \times D}$ , we construct the empirical design matrix  $\Phi = [T_0(X), \dots, T_K(X)]$ . The optimal parameter matrix  $W^*$  is derived analytically via regularized normal equations:

$$W^* = (\Phi^T \Phi + \lambda I)^{-1} \Phi^T Y_{centered} \quad (3)$$

**Definition 2** (Idempotent Subspace Projector). *Computing the compact SVD  $\Phi = U \Sigma V^T$ , we construct the rank- $r$  orthogonal projector:*

$$\Pi = V_r V_r^T \quad \text{where} \quad \Pi^2 = (V_r V_r^T)(V_r V_r^T) = V_r I_r V_r^T = \Pi \quad (4)$$

*Projecting the solved weight matrix  $W_{idem} = \Pi W^*$  guarantees that repeated application resides stably on the empirical data manifold without scale explosion [2, 3].*

### 4 In-Situ Weight Surgery on LLMs

To perform weight surgery on pre-trained LLMs without catastrophic drift across deep layers, we introduce the **Residual Polynomial Adapter (ResPoly)**:

$$\text{ResPoly}(x) = \text{MLP}_{orig}(x) + \alpha \sum_{k=0}^K \mathcal{C}_k T_k(x) \quad (5)$$

The polynomial tensor is solved directly on the residual error  $E = Y - \text{MLP}_{orig}(X)$ , guaranteeing strictly monotonic error reduction during progressive layer-by-layer surgery.

Table 1: Empirical Comparison: AdamW vs. Closed-Form Solver

| Metric                     | AdamW (150 Ep.)       | Closed-Form (Ours)                      |
|----------------------------|-----------------------|-----------------------------------------|
| Solver Execution Time      | 33.51 s               | <b>139.1 ms</b>                         |
| Speedup Factor             | 1.0×                  | <b>240.9×</b>                           |
| Reconstruction MSE         | 0.0172                | <b>0.0166</b>                           |
| Relative Idempotency Error | $1.82 \times 10^{-1}$ | <b><math>4.70 \times 10^{-5}</math></b> |
| Gradient Operations        | Yes (Multi pass)      | <b>Zero Backprop</b>                    |

Table 2: Empirical Weight Surgery on SmolLM2-135M

| Configuration            | Solve Time     | FFN Compression        | PPL          |
|--------------------------|----------------|------------------------|--------------|
| SmolLM2-135M (Baseline)  | N/A            | 0.0% (Reference)       | <b>51.53</b> |
| 1-Layer Surgery (L15)    | <b>56.2 ms</b> | <b>-50.0% (-1.33M)</b> | 55.45        |
| 2-Layer Surgery (L14-15) | 206.9 ms       | <b>-50.0% (-2.65M)</b> | 60.38        |
| 3-Layer Surgery (L13-15) | 219.2 ms       | <b>-50.0% (-3.98M)</b> | 59.42        |
| 1-Layer ResPoly (L15)    | 146.9 ms       | Residual Bit-Match     | <b>51.53</b> |

### 5 PolyFormer Architecture

For training from scratch, we define **PolyFormerBlock**, which incorporates orthogonal Chebyshev kernel attention:

$$\text{Attn}(Q, K, V) = \text{softmax} \left( \sum_{k=0}^K w_k T_k \left( \frac{QK^T}{\sqrt{d}} \right) \right) V \quad (6)$$

and meta-polynomial depth routing for adaptive early exiting (**forward\_adaptive**). While Kolmogorov-Arnold Networks (KAN) [10] introduce splines on edges, PolyFormer preserves high-speed tensorized contractions while halving FFN parameter count.

### 6 Conclusion

Orthogonal polynomial tensors and zero-backpropagation idempotent solving fundamentally decouple neural adaptation from iterative gradient descent. By enabling live in-situ model surgery with 50% FFN compression and 240× faster convergence, this approach provides a scalable, algebraically exact path for resource-constrained edge and datacenter inference [11].

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